

DEVICE TEST

1) Explain behaviour of hub device.

- * A hub is used to connect multiple ^{network} devices.
- * It is a Layer 1 device.
- * It is a broadcast device.

It's behavior:

While data send PC₁ to PC₃ it's send PC₁ is send's electrical signal to the hub the hub is doesn't check the destination MAC it's simply copies/forward the signal to every other ports. so data was broadcast so collision occurred.

2) Explain behaviour of switch device.

- * A switch is used to connect multiple n/w device.
- * A switch is a intelligent device.
- * It's a OSI Layer 2 device, and use MAC address to forward Ethernet frame.

It's behavior:

While data send PC₁ to PC₂ when switch receive the data it first learn the source MAC address and store. MAC table if destination MAC address doesn't at MAC table switch is broadcast the data, after identify destination MAC address store the MAC table. again PC₂ → PC₁ data was send already destination MAC was know so the switch is unicast the data.

3. Explain behaviour of Router device.

- * Router used to break the broadcast domain
- * It's a Layer 3 device
- * It's contain a routing table
- * It's used to connect different

network.
* It's a
behaviour!

unicast device.

When a router receive the data
It's check the destination IP address
If the destination network route is
available at routing table packets
through the destination network If route
is not available router will drop the
packets.